

**САНКТ-ПЕТЕРБУРГСКИЙ НАЦИОНАЛЬНЫЙ
ИССЛЕДОВАТЕЛЬСКИЙ УНИВЕРСИТЕТ ИТМО**

Дисциплина: Бэк-энд разработка

Отчет

Лабораторная работа №3
“Контейнеризация написанного приложения средствами
docker”

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Задача

- Реализовать Dockerfile для каждого сервиса.
- Написать общий docker-compose.yml.
- Настроить сетевое взаимодействие между сервисами.

Ход работы

Теоретические сведения

- Docker — изоляция приложения с зависимостями в образе.
- Docker Compose — декларативное описание multi-container приложения: сервисы, сети, volumes, переменные окружения, depends_on и healthcheck.

В проекте используется bridge-сеть rental-net: контейнеры обращаются друг к другу по DNS-имени сервиса (auth-service, postgres-auth, ...).

Dockerfile для каждого сервиса

Общий подход (multi-stage build)

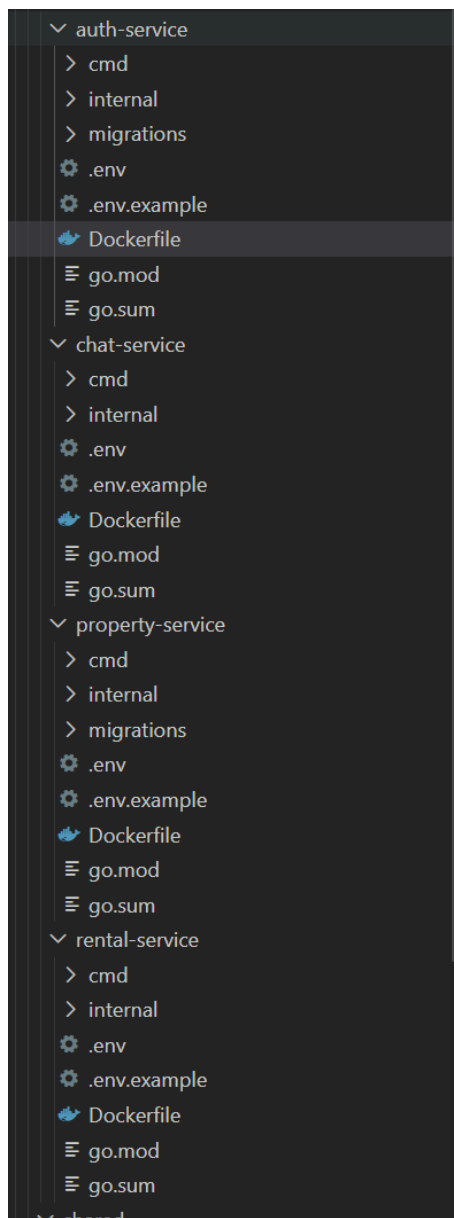
1. builder — golang:1.23-alpine, копирование go.work, shared/, services/<name>/, go build
2. runtime — alpine:3.20, бинарник, EXPOSE порта сервиса

Пример (auth-service):

```
1 FROM golang:1.23-alpine AS builder
2
3 WORKDIR /src
4 COPY shared ./shared
5
6 COPY services/auth-service ./services/auth-service
7
8 WORKDIR /src/services/auth-service
9
10 RUN go mod download
11
12 RUN CGO_ENABLED=0 GOOS=linux go build -o /auth-service ./cmd
13
14
15 FROM alpine:3.20
16 RUN apk add --no-cache ca-certificates
17
18 WORKDIR /app
19
20 COPY --from=builder /auth-service .
21
22 EXPOSE 8081
23 CMD ["/auth-service"]
```

Аналогичные Dockerfile:

- services/auth-service/Dockerfile → порт 8081
- services/property-service/Dockerfile → 8082
- services/rental-service/Dockerfile → 8083
- services/chat-service/Dockerfile → 8084



☐		deploy	-	-	-	1.27%	2 hours ago			
☐		postgres-rental-1	29f724b46101	postgres:16-alpine		0%	6 hours ago			
☐		rabbitmq-1	608592bdd8f8	rabbitmq:3.13-management	15672:15672	0.32%	6 hours ago			
☐		postgres-chat-1	762ec53b10eb	postgres:16-alpine		0.01%	6 hours ago			
☐		postgres-auth-1	035d4d757666	postgres:16-alpine		0.9%	6 hours ago			
☐		postgres-property-1	572d41e107fe	postgres:16-alpine		0.01%	6 hours ago			
☐		property-service-1	5cd14d6feca8	deploy-property-service		0%	6 hours ago			
☐		auth-service-1	1264baee4c9c	deploy-auth-service		0%	6 hours ago			
☐		chat-service-1	1e25a9d5408d	deploy-chat-service		0%	6 hours ago			
☐		api-gateway-1	9cff10e60bfd	nginx:1.27-alpine	8080:80	0%	6 hours ago			
☐		rental-service-1	436b47b323c4	deploy-rental-service		0.03%	2 hours ago			

Docker Compose

Файл: labs/rental-platform/deploy/docker-compose.yml

Сервисы инфраструктуры

Сервис	Образ	Назначение
postgres-auth	postgres:16-alpine	БД auth_db
postgres-property	postgres:16-alpine	БД property_db
postgres-rental	postgres:16-alpine	БД rental_db
postgres-chat	postgres:16-alpine	БД chat_db
rabbitmq	rabbitmq:3.13-management-alpine	Очереди + UI :15672

Для каждой БД — отдельный volume (auth_data, ...).

Прикладные сервисы

Переменные окружения задают подключение внутри сети Docker:

yaml

DB_URL: postgres://auth:auth@postgres-auth:5432/auth_db?sslmode=disable

AUTH_SERVICE_URL: http://auth-service:8081

PROPERTY_SERVICE_URL: http://property-service:8082

RABBITMQ_URL: amqp://guest:guest@rabbitmq:5672/

JWT_SECRET: dev_secret_change_me

depends_on с condition: service_healthy для Postgres и RabbitMQ.

API Gateway

Сервис api-gateway (nginx:1.27-alpine):

- Порт хоста 8080:80
- Volumes: infra/nginx/nginx.conf, infra/nginx/conf.d/api-gateway.conf

▷Run Service

api-gateway:

image: nginx:1.27-alpine

ports:

- "8080:80"

volumes:

- ../infra/nginx/nginx.conf:/etc/nginx/nginx.conf:ro

- ../infra/nginx/conf.d:/etc/nginx/conf.d:ro

depends_on:

- auth-service

- property-service

- rental-service

- chat-service

networks:

- rental-net

networks:

rental-net:

driver: bridge

volumes:

auth_data:

property_data:

rental_data:

chat_data:

```

1 services:
2   >Run Service
3   postgres-auth:
4     image: postgres:16-alpine
5     environment:
6       POSTGRES_USER: auth
7       POSTGRES_PASSWORD: auth
8       POSTGRES_DB: auth_db
9     volumes:
10       - auth_data:/var/lib/postgresql/data
11     networks:
12       - rental-net
13     healthcheck:
14       test: ["CMD-SHELL", "pg_isready -U auth -d auth_db"]
15       interval: 5s
16       timeout: 5s
17       retries: 5
18
19   >Run Service
20   postgres-property:
21     image: postgres:16-alpine
22     environment:
23       POSTGRES_USER: property
24       POSTGRES_PASSWORD: property
25       POSTGRES_DB: property_db
26     volumes:
27       - property_data:/var/lib/postgresql/data
28     networks:
29       - rental-net
30     healthcheck:
31       test: ["CMD-SHELL", "pg_isready -U property -d property_db"]
32       interval: 5s
33       timeout: 5s
34       retries: 5
35
36   >Run Service
37   postgres-rental:
38     image: postgres:16-alpine
39     environment:
40       POSTGRES_USER: rental
41       POSTGRES_PASSWORD: rental
42       POSTGRES_DB: rental_db
43     volumes:
44       - rental_data:/var/lib/postgresql/data
45     networks:
46       - rental-net
47     healthcheck:
48       test: ["CMD-SHELL", "pg_isready -U rental -d rental_db"]
49       interval: 5s
50       timeout: 5s
51       retries: 5

```

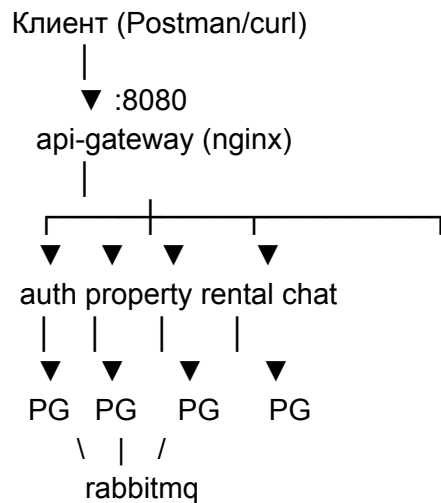
```

[+] Running 10/10
 ✓ auth-service           Built           0.0s
 ✓ property-service       Built           0.0s
 ✓ rental-service         Built           0.0s
 ✓ Container deploy-postgres-rental-1 Healthy        11.9s
 ✓ Container deploy-postgres-auth-1  Healthy        10.9s
 ✓ Container deploy-postgres-property-1 Healthy        11.4s
 ✓ Container deploy-rabbitmq-1       Healthy        11.9s
 ✓ Container deploy-auth-service-1   Running        0.0s
 ✓ Container deploy-property-service-1 Running        0.0s
 ✓ Container deploy-rental-service-1 Started        12.2s
PS D:\itmo-backend-6sem\ITMO-ACS-Backend-2026-A\БП1.1\Данилова Анастасия\labs\rental-platform>

```

Сетевое взаимодействие

Схема



Все контейнеры в сети rental-net (driver: bridge).

Проверка связности

Из хоста:

bash

curl http://localhost:8080/health

curl http://localhost:8080/api/properties

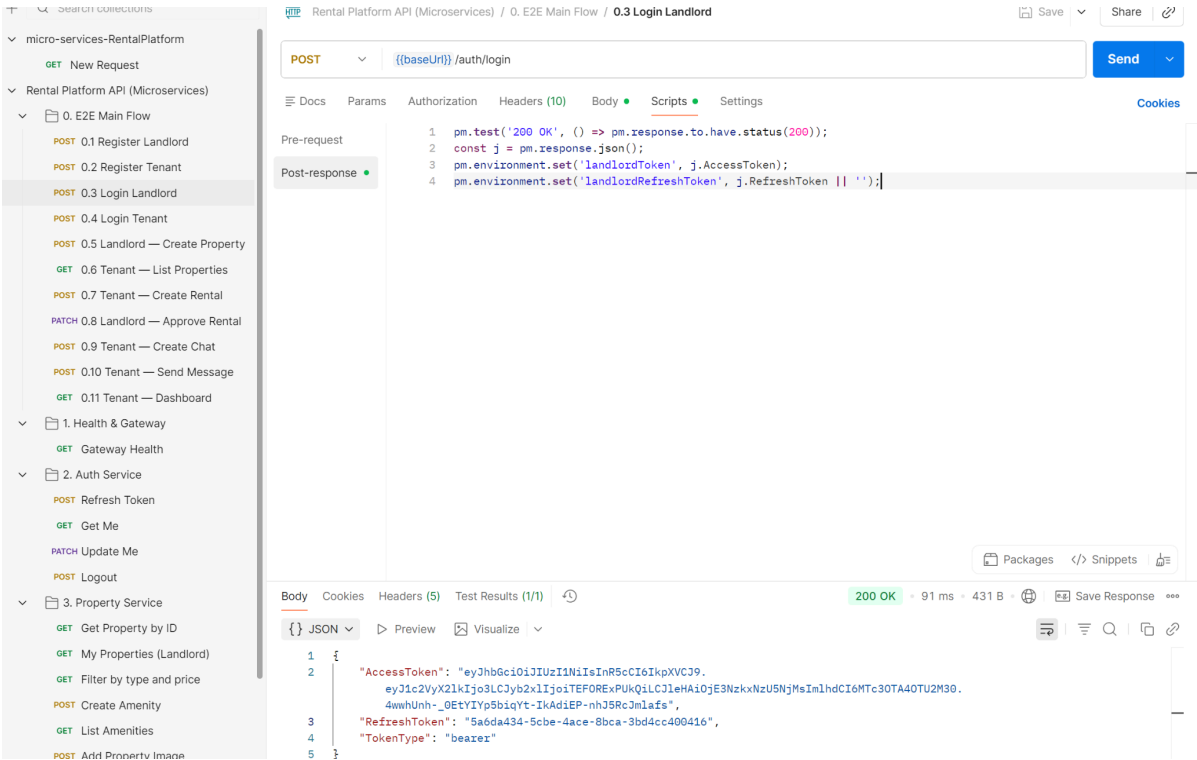
```

andan@NASTUSHKA MINGW64 /d/itmo-backend-6sem/ITMO-ACS-Backend-2026-A/БР1.1/Данилова Анастасия (main)
$ curl http://localhost:8080/api/properties
[{"id":3,"title":"Cozy apartment near metro","city":"Saint Petersburg","price_per_month":65000,"is_available":true},{"id":2,"title":"Cozy apartment near metro","city":"Saint Petersburg","price_per_month":65000,"is_available":true},{"id":1,"title":"Cozy apartment near metro","city":"Saint Petersburg","price_per_month":65000,"is_available":true}]
andan@NASTUSHKA MINGW64 /d/itmo-backend-6sem/ITMO-ACS-Backend-2026-A/БР1.1/Данилова Анастасия (main)
$ curl http://localhost:8080/health
{"status":"ok","gateway":true}
andan@NASTUSHKA MINGW64 /d/itmo-backend-6sem/ITMO-ACS-Backend-2026-A/БР1.1/Данилова Анастасия (main)
$ ;3R;6R

```

Между сервисами (имена DNS из compose):

- rental → http://property-service:8082/internal/properties/1
- rental → http://auth-service:8081/internal/users/validate



Порты на хосте

Порт	Сервис
8080	API Gateway
5672, 15672	RabbitMQ (+ Management UI)

Порты 8081–8084 не проброшены наружу — доступ только через gateway.

Queues

All queues (1)

Pagination

Page 1 of 1 - Filter: ☐ Regex ?

Overview					Messages			Message rates			+/-	
Virtual host	Name	Type	Features	State	Ready	Unacked	Total	incoming	deliver /	get /	ack	
/	rental-service.user-deleted	classic	D	running	0	0	0					

Add a new queue

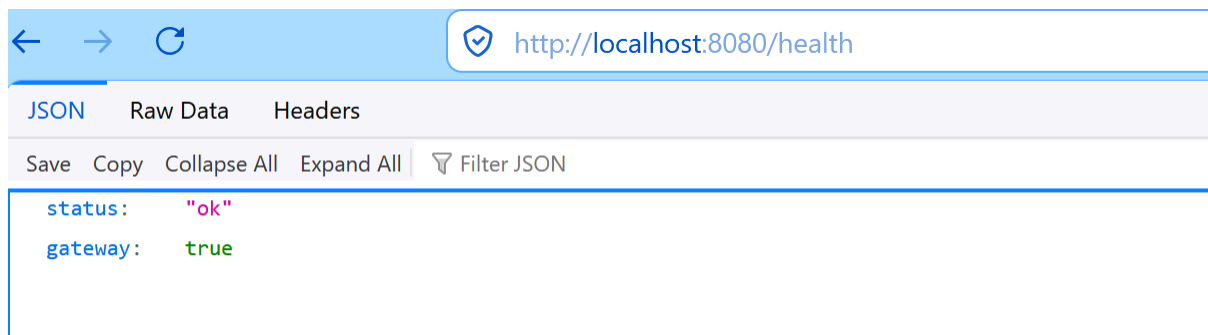
Запуск и Makefile

```
bash
cd labs/rental-platform
make docker-up # build + up -d
make docker-down
make docker-logs
```

```
andan@Anastasias-MacBook-Air rental-platform % make docker-build
Makefile:27: warning: overriding commands for target `docker-build'
Makefile:24: warning: ignoring old commands for target `docker-build'
Makefile:30: warning: overriding commands for target `docker-build'
Makefile:27: warning: ignoring old commands for target `docker-build'
Makefile:33: warning: overriding commands for target `docker-build'
Makefile:30: warning: ignoring old commands for target `docker-build'
Makefile:36: warning: overriding commands for target `docker-build'
Makefile:33: warning: ignoring old commands for target `docker-build'
docker compose -f deploy/docker-compose.yml up -d --build rental-service
[+] Building 162.6s (15/35)
=> [internal] load local bake definitions 0.0s
=> => reading from stdin 2.15kB 0.0s
=> [rental-service internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 436B 0.0s
=> [property-service internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 448B 0.0s
=> [auth-service internal] load build definition from Dockerfile 0.1s
=> => transferring dockerfile: 424B 0.0s
=> [auth-service internal] load metadata for docker.io/library/golang:1.23-alpine 140.3s
=> [property-service internal] load metadata for docker.io/library/alpine:3.20 139.0s
=> [auth] library/golang:pull token for registry-1.docker.io 0.0s
=> [auth] library/alpine:pull token for registry-1.docker.io 0.0s
=> [property-service internal] load .dockerignore 0.0s
=> => transferring context: 2B 0.0s
=> [property-service stage-1 1/4] FROM docker.io/library/alpine:3.20@sha256:d9e853e87e55526f6b2917df91a2115c36dd7c696a35be 0.1s
=> => resolve docker.io/library/alpine:3.20@sha256:d9e853e87e55526f6b2917df91a2115c36dd7c696a35be12163d44e6e2a4b6bc 0.0s
=> [auth-service internal] load build context 0.1s
=> => transferring context: 36.61kB 0.1s
=> [property-service builder 1/7] FROM docker.io/library/golang:1.23-alpine@sha256:383395b794dffa5b53012a212365d40c8e3710 20.8s
=> => resolve docker.io/library/golang:1.23-alpine@sha256:383395b794dffa5b53012a212365d40c8e37109a626ca30d6151c8348d380b5f 0.0s
=> => sha256:d3178a7b27090e22bc2464346d1cdc8f59d0fd7a241452e8148342edf3d0b1e9 126B / 126B 0.3s
=> => sha256:d5791340ef181f05ef1e99760e4be81cb2008d99b2bc1e56b2faa9b6295d0ac3 55.57MB / 74.07MB 20.4s
=> => sha256:8371a51cbc44c1323f96695eaeaebe1c44b2d30d138a279b348af5c18b2c5e24 282.45kB / 282.45kB 0.9s
=> [property-service] resolving provenance for metadata file 0.1s
[+] up 14/14
✓ Image deploy-auth-service Built 447.8s
✓ Image deploy-rental-service Built 447.8s
✓ Image deploy-property-service Built 447.8s
✓ Network deploy_rental-net Created 0.2s
✓ Volume deploy_property_data Created 0.1s
✓ Volume deploy_auth_data Created 0.1s
✓ Volume deploy_rental_data Created 0.0s
✓ Container deploy-rabbitmq-1 Healthy 22.0s
✓ Container deploy-postgres-rental-1 Healthy 22.0s
✓ Container deploy-postgres-auth-1 Healthy 17.6s
✓ Container deploy-postgres-property-1 Healthy 21.1s
✓ Container deploy-auth-service-1 Created 1.0s
✓ Container deploy-property-service-1 Created 0.7s
✓ Container deploy-rental-service-1 Created 0.6s
```

Health checks

Каждый микросервис: GET /health → {"status":"ok","service":"..."}



Выводы

1. Для каждого из 4 микросервисов создан multi-stage Dockerfile с минимальным runtime-образом Alpine.
2. docker-compose.yml объединяет приложения, 4 PostgreSQL, RabbitMQ и nginx gateway.
3. Сетевое взаимодействие настроено через Docker network и DNS-имена сервисов; клиенты используют единую точку входа :8080.
4. Volumes обеспечивают сохранность данных БД между перезапусками.
5. Стек готов к подключению CI/CD (сборка образов, compose up в pipeline).